# FROM EXCEL TO R



SUND DATA LAB

# WHO ARE WE?

Center for Health Data Science (**HeaDS**):

**Read more on HeaDS Website!**
https://heads.ku.dk



Chelsea



Diana



Suze

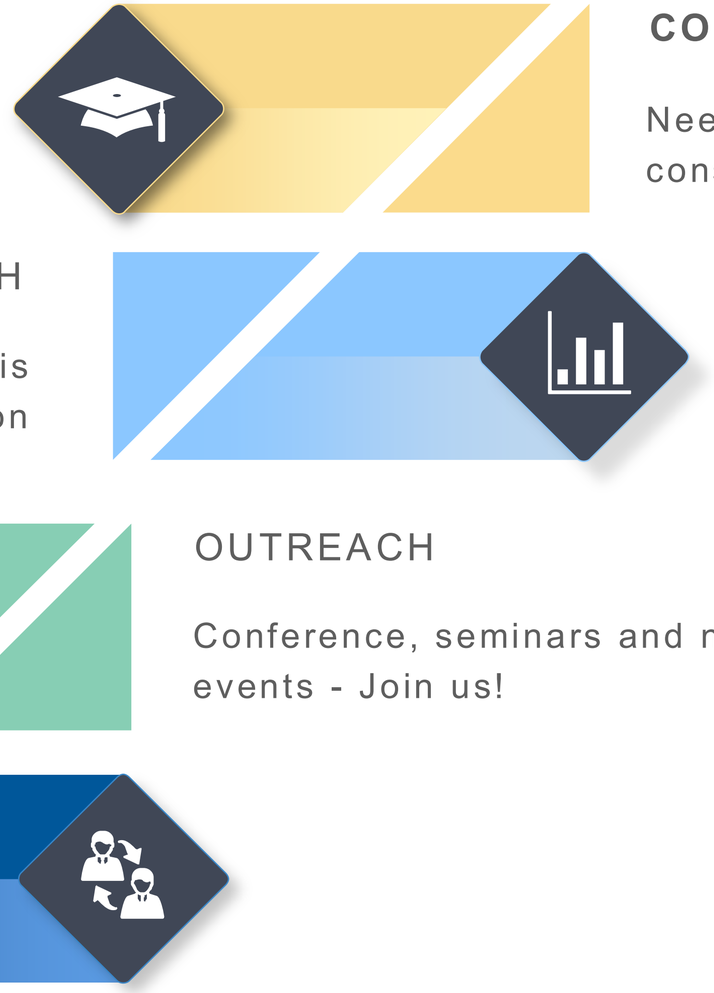
**Follow HeaDS on LinkedIn!**
https://www.linkedin.com/company/ucph-heads/

*Data Science Laboratory (DSL) - https://datalab.science.ku.dk/*

UCPH HeaDS

SUND DATA LAB

# WHAT WE DO?



## CONSULTATIONS

Need guidance? Drop by for a consultation on your DS analysis

## **COMMISSIONED** RESEARCH

Commissioned DS Analysis
Commissioned Supervision

## OUTREACH

Conference, seminars and networking events - Join us!

## COURSES & WORKSHOPS

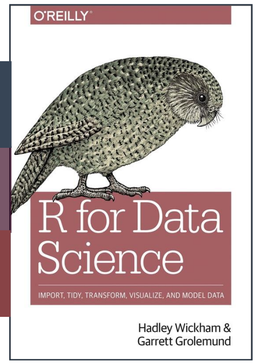
Data Science skills, Tools and HDS Topics

# THE PRACTICALS

Two days: 9.00-16.00. There will be coffee breaks, we promise

“R for Data Science” - a generally useful book on R, also for this course

The course is build on hands-on presentations & exercises



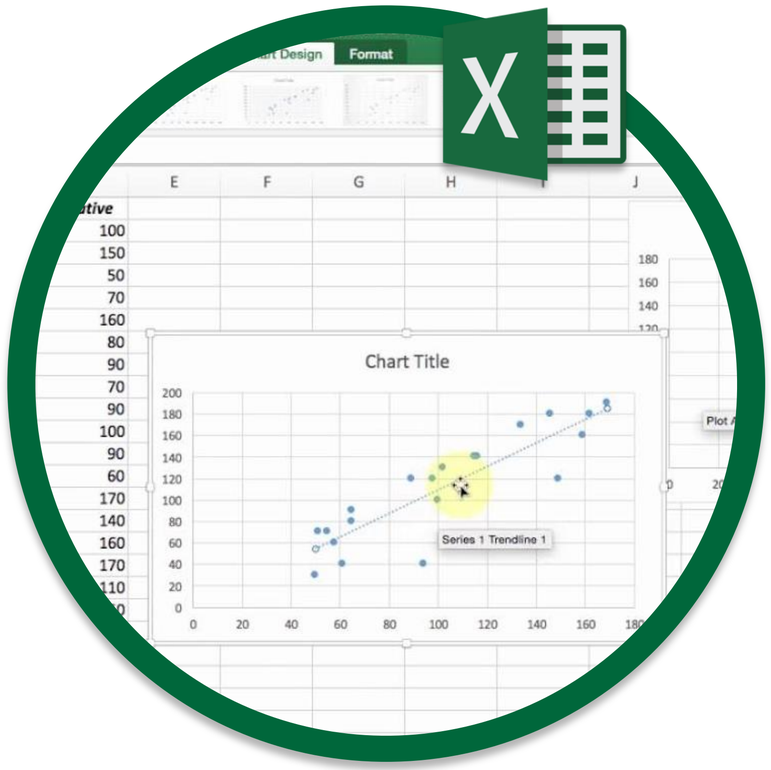

Download and install the newest version of R (https://cran.r-project.org/)

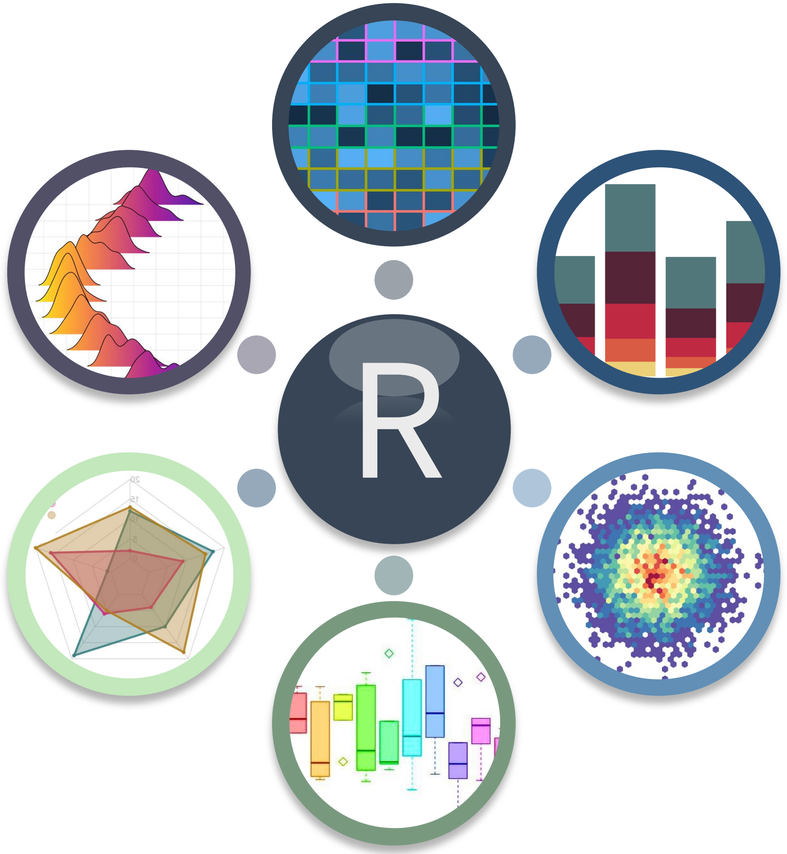
Download and install the newest version of R-studio (https://posit.co/download/rstudio-desktop/)

Go to course website: https://center-for-health-data-science.github.io/FromExceltoR/

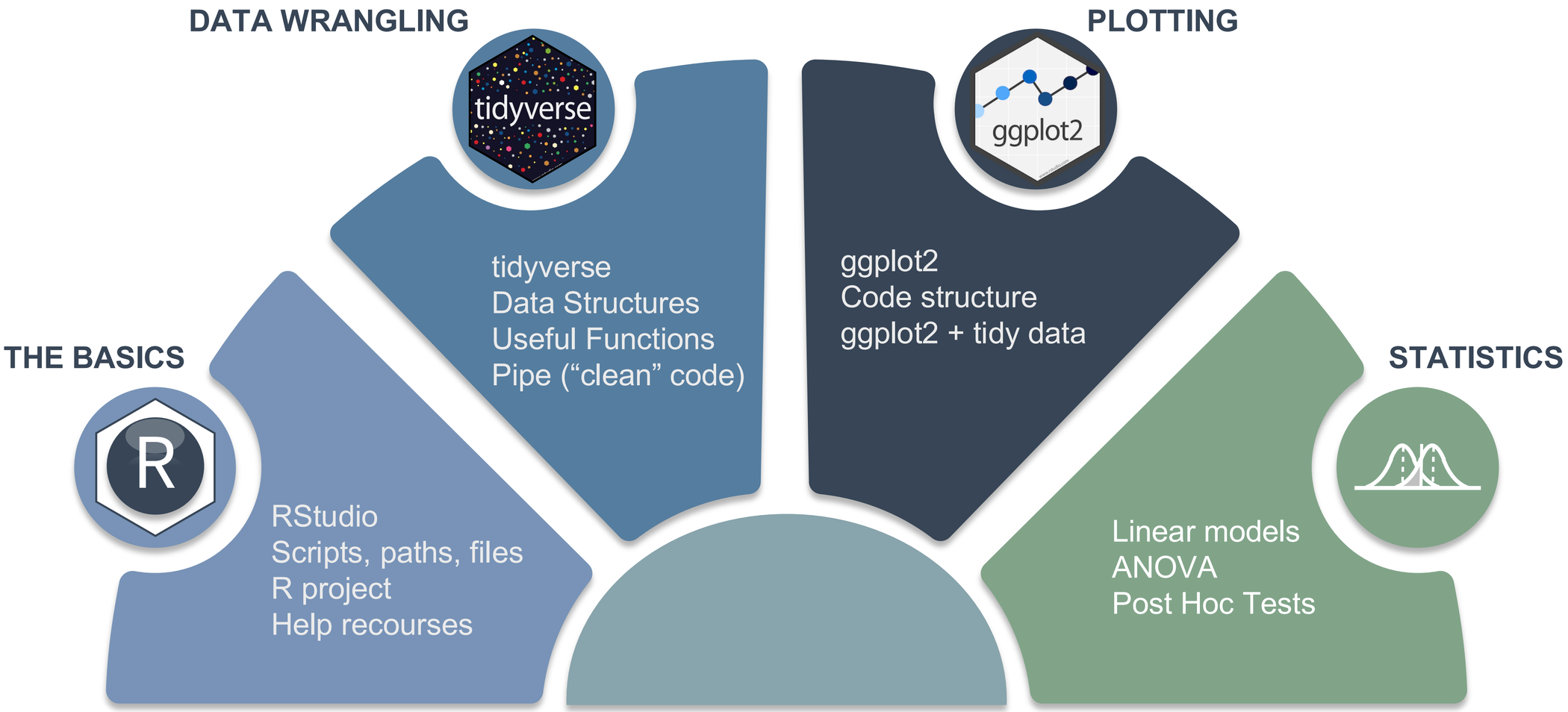
WELCOME TO

# FROM EXCEL TO R



# WHAT WILL I LEARN?



# PROGRAM

**DAY 1**

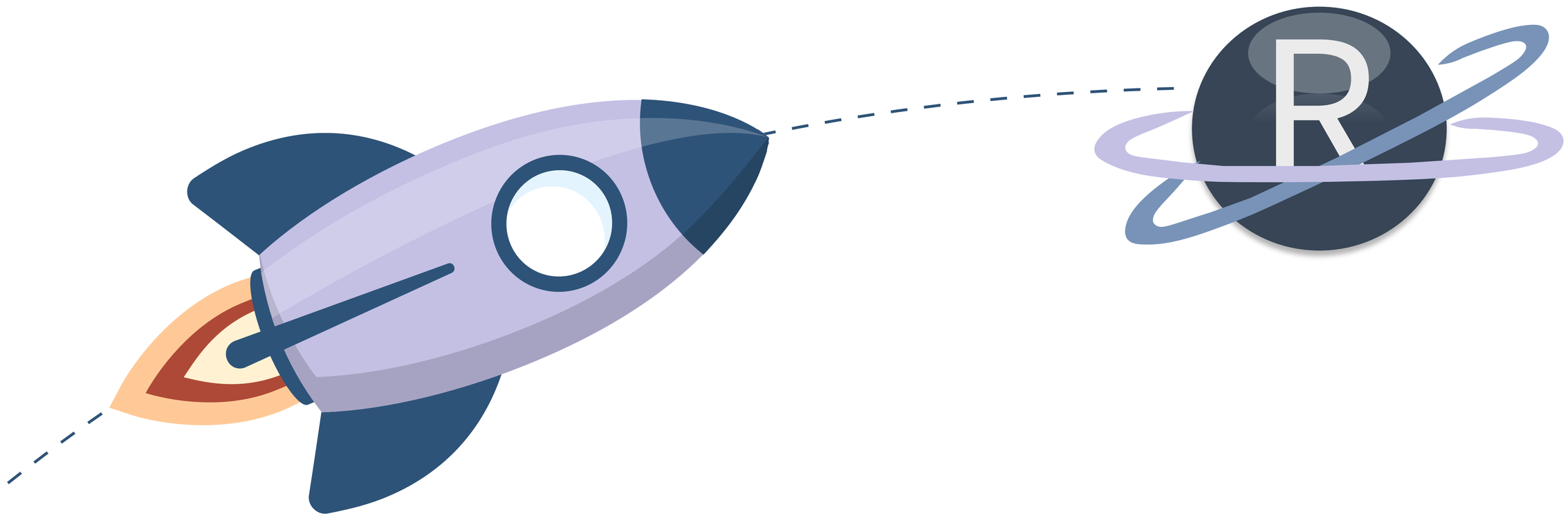
08:45 - Installation Issues + Coffee
09:00 - Introduction to R
09:45 - Exercise 0
10:15 - Break
10:30 - Introduction to Quarto
11:10 - Exercise 1
12:00 - Lunch
13:00 – Tidyverse
14:00 - Break
14:15 - Tidyverse Exercise 2
16:00 - See you tomorrow

**DAY 2**

08:45 - Coffee + Optional Q&A
09:00 - ggplot2 + Exercise 3
10:30 - Break
10:45 - ggplot2 Exercise 3
12:00 - Lunch
13:00 - Applied Statistics
14:00 - Break
14:15 - Statistics Exercise
15:00 - Cool things in R + Course Evaluation
15:15 - DataLabs Services / Where to get Help
16:00 - Bye Bye Bye!

FROM EXCEL TO R

# LET'S GET STARTED



# R & FRIENDS

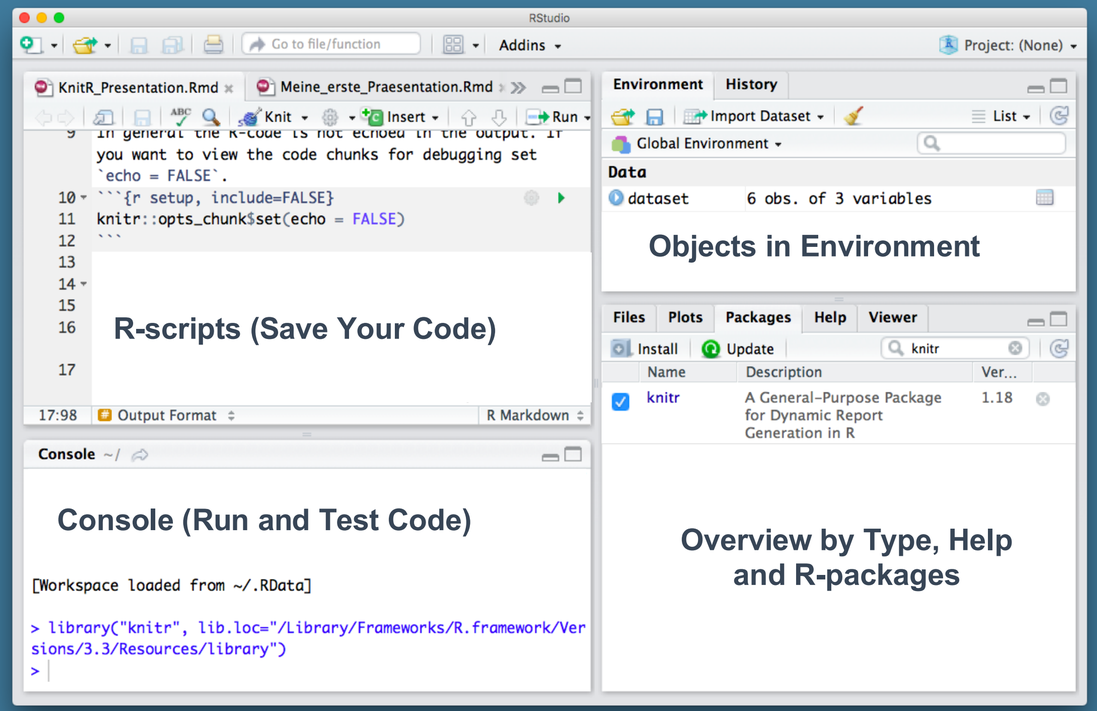


**Scripting / Programming Language**




**Reports (html, pdf, word)**




**R Code Interpreter and Editor**

# THE ANATOMY OF R

The **programming language** R may be used from within in the **Code Editor** RStudio

The R code you write may be saved in **scripts** with extensions: .R .Rmd .qmd

**Functions** perform actions based on **input arguments** and return an output:

```
mean(x)
sd(x)
```

**Packages** are bundles/libraries of functions.




```
select()
mutate()
```

```
tibble()
```

https://www.quarto.org/

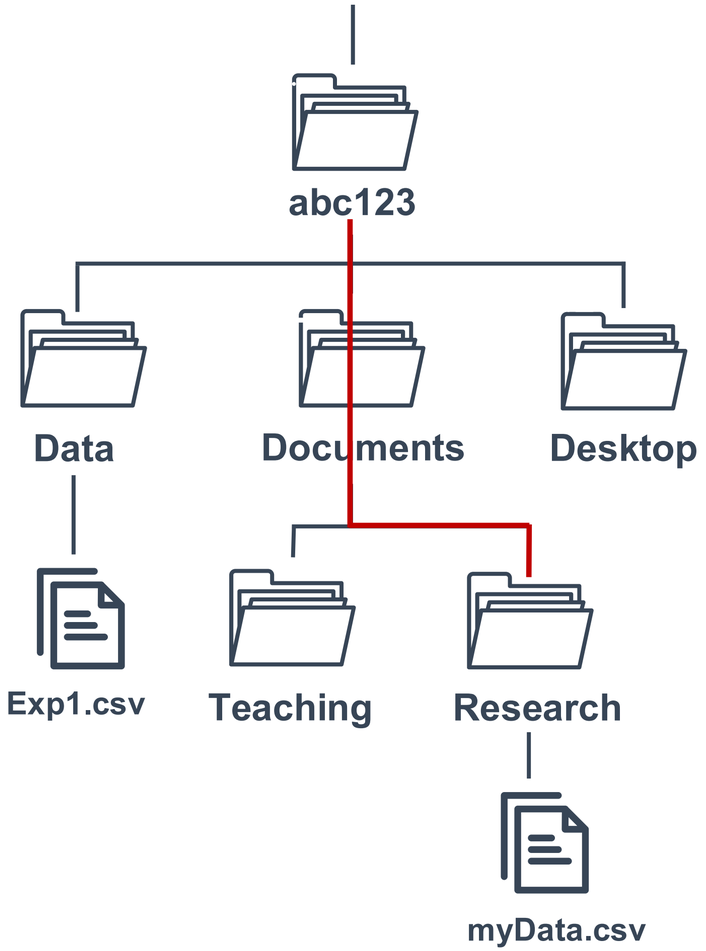
# PATHS – LOCATION, LOCATION, LOCATION

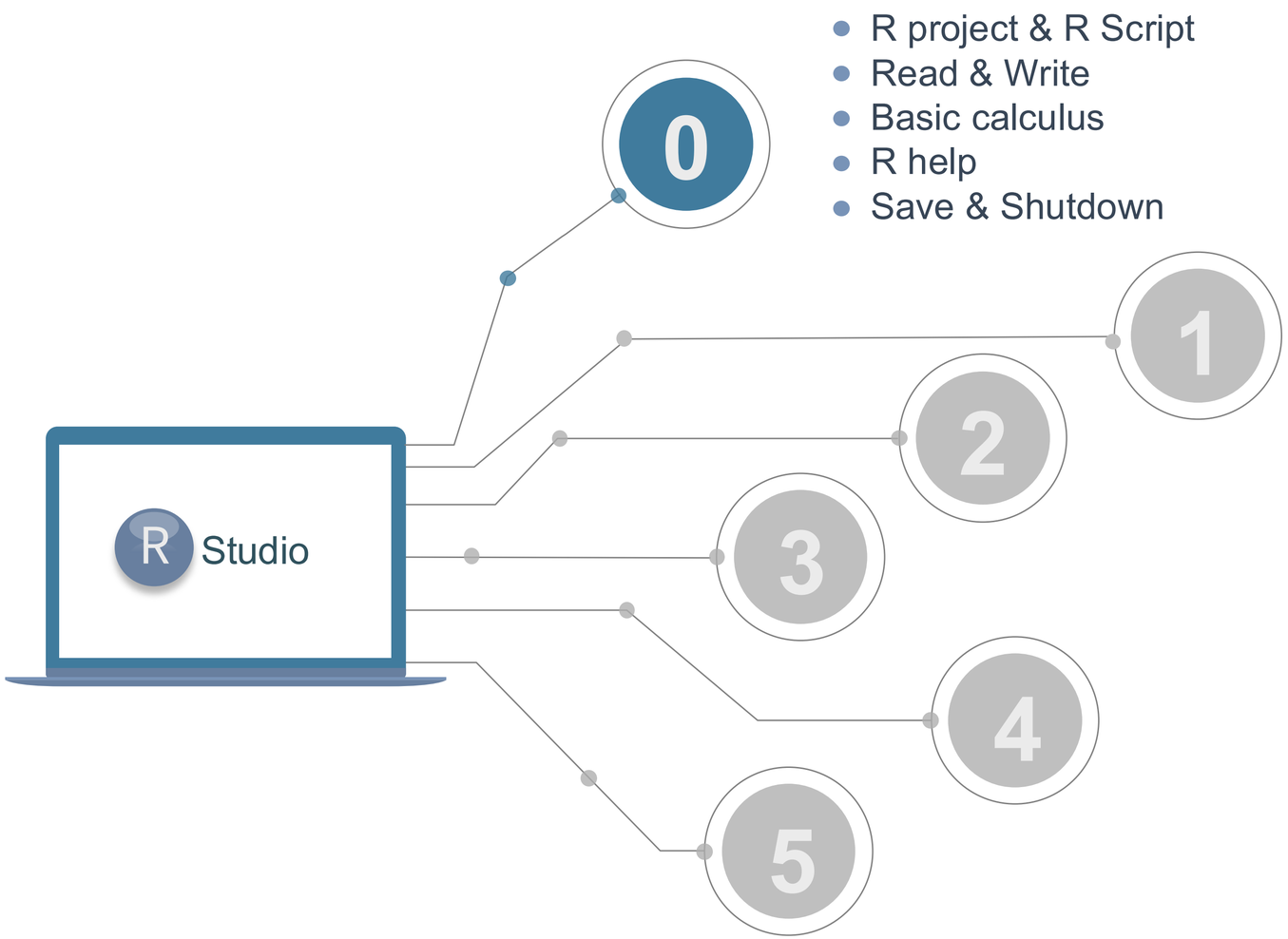
- The directories on your computer are organized in a **File Tree**
- **Move** between directories or **point to** a place in the tree with a specific file

We do this by specifying a **path:**

- **Move to:** `/abc123/Documents/Research`
- **Point to:** `/abc123/Documents/Research/myData.csv`

**Go through Exercise 0 in plenum!**



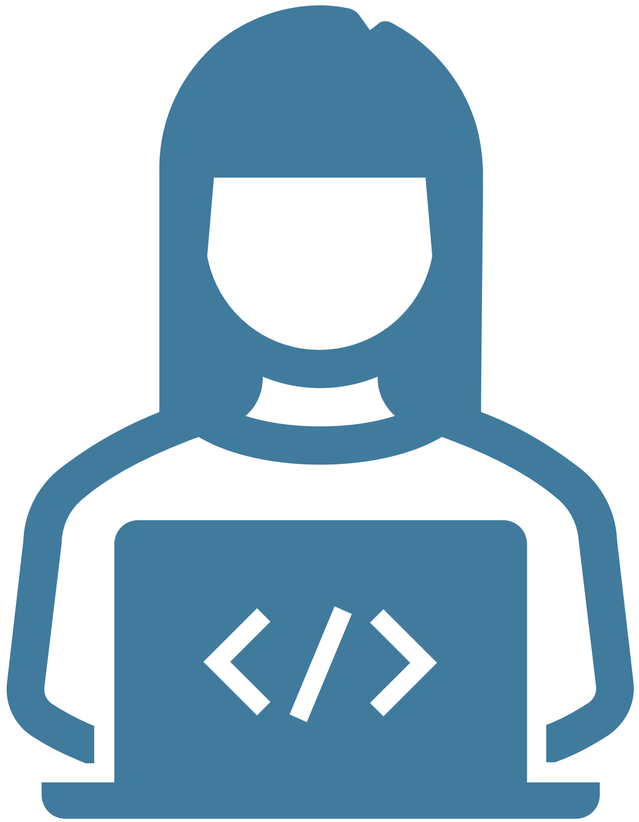



FUNDAMENTALS

# EXERCISE **0**

FROM EXCEL TO R

# LIVE CODING 0 – INTRO TO R



# ONLINE RESOURCES FOR **R**

https://www.r-project.org/

## GET STARTED

https://rseek.org/

https://rstudio.com/resources/cheatsheets/

http://www.cookbook-r.com/

https://www.statmethods.net/r-tutorial/index.html

## GRAPHICS

https://www.r-graph-gallery.com/

http://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html

## BOOKS & COURSES

https://www.r-bloggers.com/best-books-to-learn-r-programming/

https://www.datacamp.com/

https://www.codecademy.com/

https://www.coursera.org/

## OTHER RESOURCES

https://github.com/trending/r

https://blog.revolutionanalytics.com/

https://stackoverflow.com/questions/tagged/r

https://www.rstudio.com/wp-content/uploads/2016/10/r-cheat-sheet-3.pdf

# INTRO CHEAT SHEET

## Basics:

```
getwd() # location
install.packages("package_name") # install packages
library(package_name) # load packages
```

## Read in Data:

```
read.xlsx("name.xlsx") # library(readxl)
read.delim("name.txt", sep ="\t")
read.csv("name.csv", sep=";")
```

GETTING STARTED

## Overview:

```
head(df, n=10), tail(df, n=10) # first or last 10 rows
unique(), table(), count() # unique vals, count vals
```

```
view() # view data as table
df$col1 # extract column from dataframe
nrow(df), ncol(df) # number of rows/columns
```

OVERVIEW

## Type/Class:

```
class() # get data type/class
is.numeric(x), is.character(), is.factor() # is data type
as.numeric(x), as.character(), as.factor() # change data type
```

## Conditions:

```
a == b   a != b   a > b   a < b
a >= b   a <= b   is.na() is.null()
```

DATA TYPES & CONDITIONS

## Summary Stats:

```
summary() # summary statistics
mean(), median(), sd(), sum(), log(), max(),
min(), quantile(), var()
```

## Plots:

```
plot(x)
plot(x, y) or plot(col1, col2, df) # scatter
hist(x) # histogram
```

STATISTICS & BASE PLOTS

https://www.quarto.org/

# QUARTO DOCUMENTS

- Tool which supports **reproducible** coding/publication (reports)
- Quarto works with a **markup language**
- A markup language is **text-coding system** which specifies structure & format of a document and the text within it
- We **render** (compile) a Quarto → html, pdf, word, website

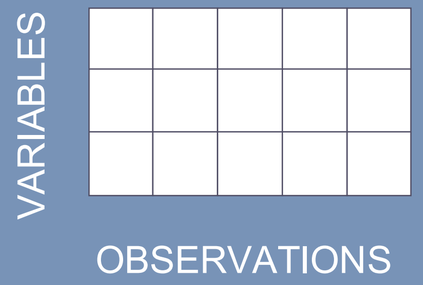
**Pointers for quarto:**

- Separate sections with headers
- Do not overfill code chunks
- Write meaningful and precise comments

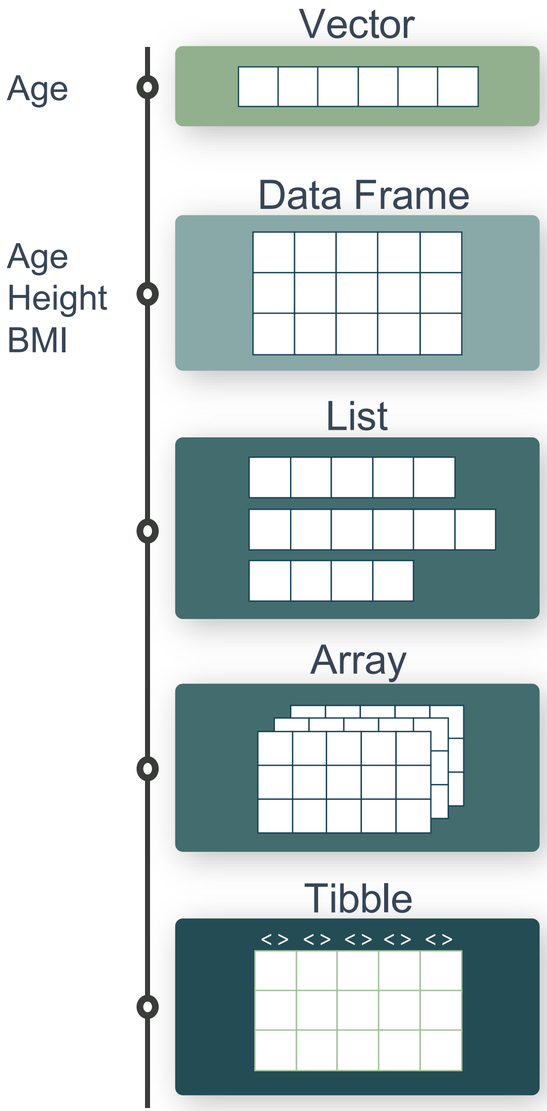


FROM EXCEL TO R

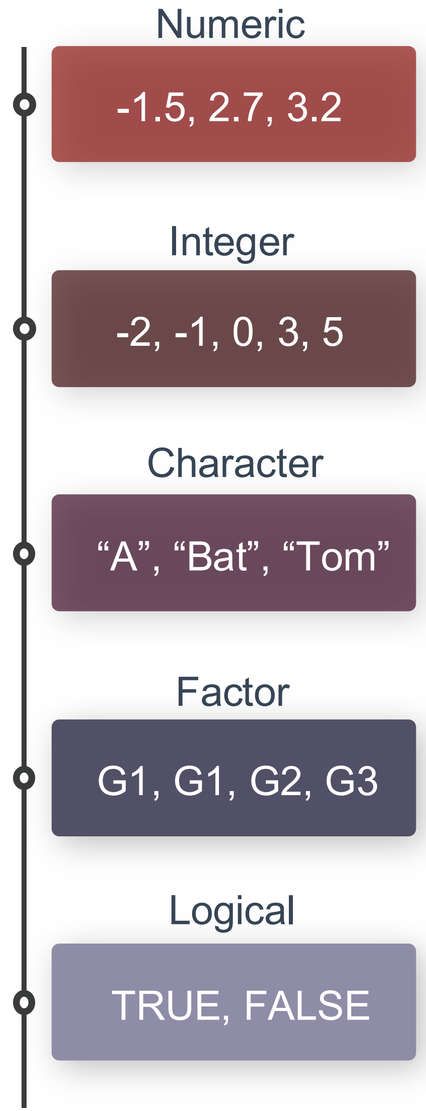
# R DATA TYPES & STRUCTURES




## DATA STRUCTURES

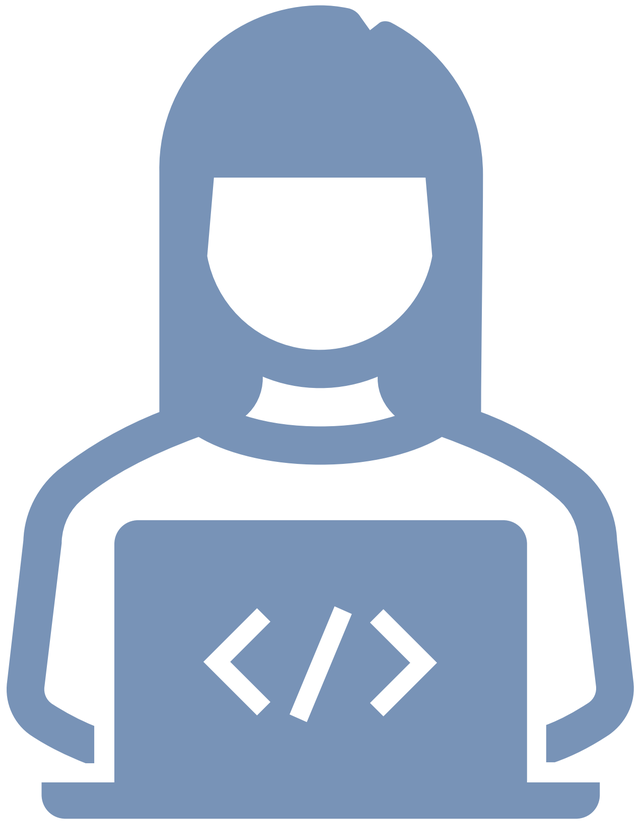



## DATA TYPES



FROM EXCEL TO R

# LIVE CODING 1 – QUARTO



https://quarto.org

# QUARTO CHEAT SHEET

## YAML parameter:

```
---
title: My Project Name
output:
      html_document (pdf_document, ...)
---
```

## Code Chunk:



**Source mode:**

```
```{r}
# some R code
```
```

**Visual mode:**

```
{r}
# some R code
```

GETTING STARTED

## Code Options:

```
{r echo = FALSE} # don't print code (default is TRUE)
{r eval = FALSE} # don't run code (default is TRUE)
{r error = FALSE} # don't display error message (default is TRUE)
(Can also be set for warning and message)
```

## Figure Options:

```
fig.align (='left', 'right', 'center')
fig.cap (= 'my figure cation')
fig.height (= n), fig.width (= n)
```

CHUNK OPTIONS

## Source mode:

### Header

```
Header size ranging from largest (one #)
to smallest (six #):
# my.text, ## my.text, ### my.text, etc.
```

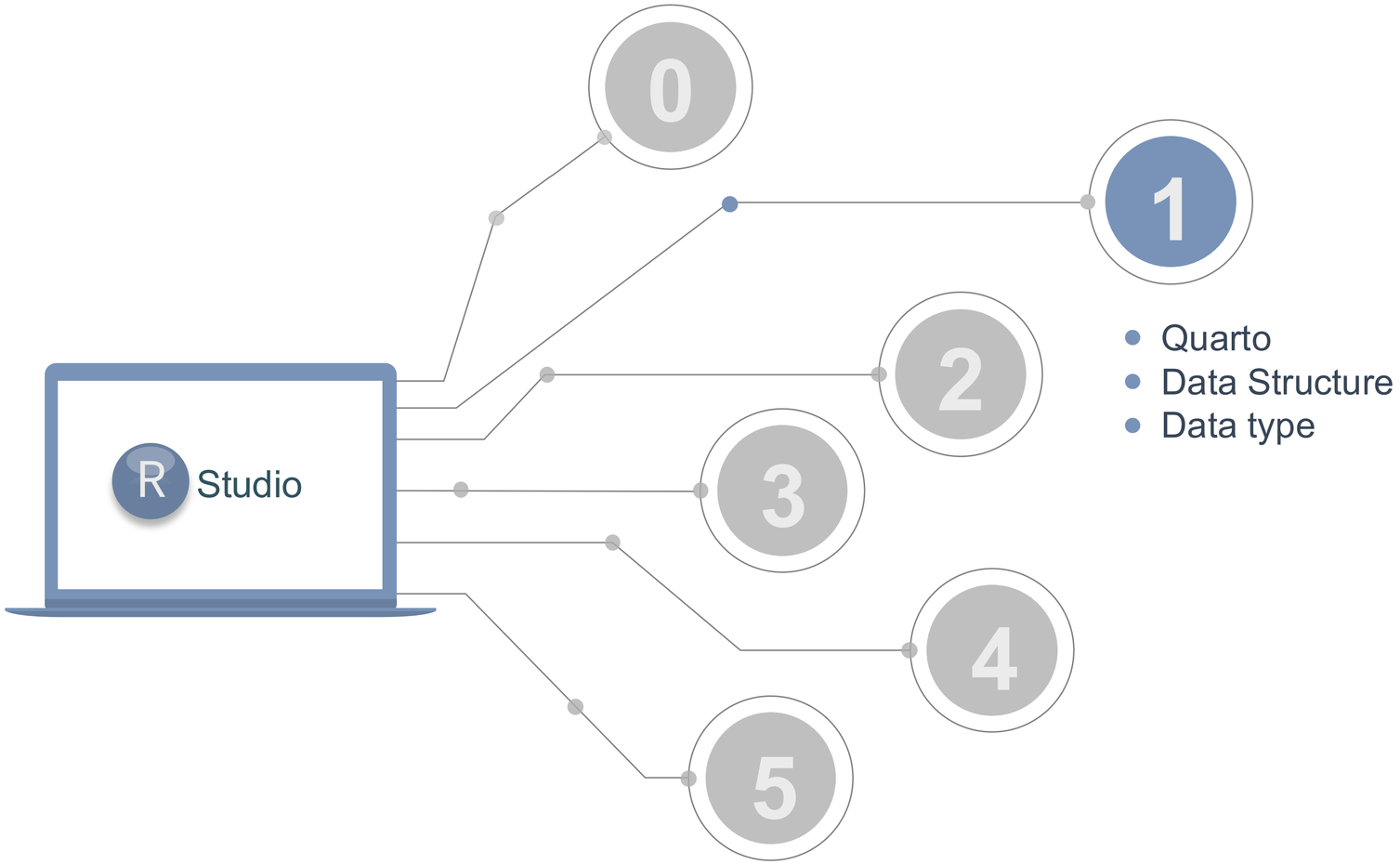
## Text

```
*italics*
**bold**
`highlighted`
```

## Lists

```
* List item1 (filled dot)
     + sub-item1 (open dot)
1. List item1 (numbered)
     i) sub-item1 (roman)
```

TEXT




QUARTO

# EXERCISE **1**

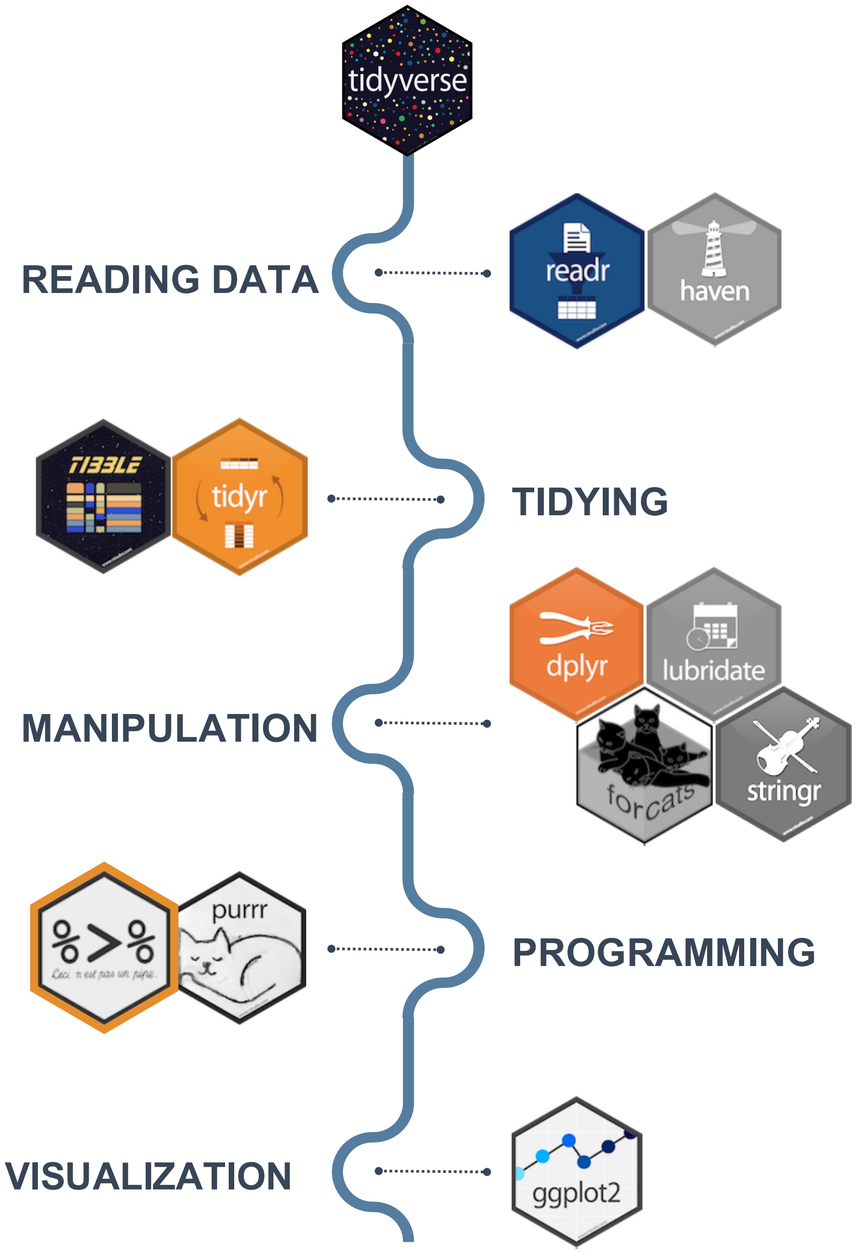
https://www.tidyverse.org/

# TIDYVERSE

tidyverse is a collection of R packages for data science

"The packages share an underlying design philosophy, grammar, and data structures." *Wickham and Grolemund*

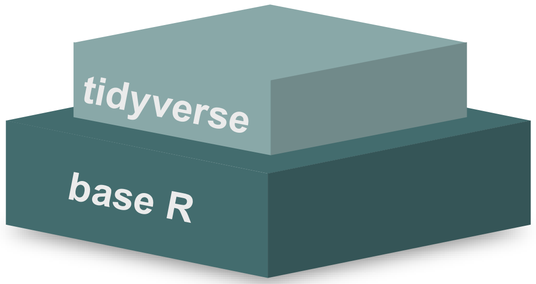
tidyverse is used to "tidy up" your datasets, so they are easy to work with



https://www.tidyverse.org/

# CECI N'EST PAS UNE PIPE %>%

- You do NOT have to “chose” between tidyverse and base R




## BENEFITS

- Short & well-organised code
- Tidy datasets, easy to work with
- Great documentation
- Functions with logical names & inputs

## CONSIDERATIONS

- Can be less stable
- “Different syntax”
- Can be computationally slow

**base R**

```
# think from the inside out

g(f(x,y),z)
```

**tidyverse**

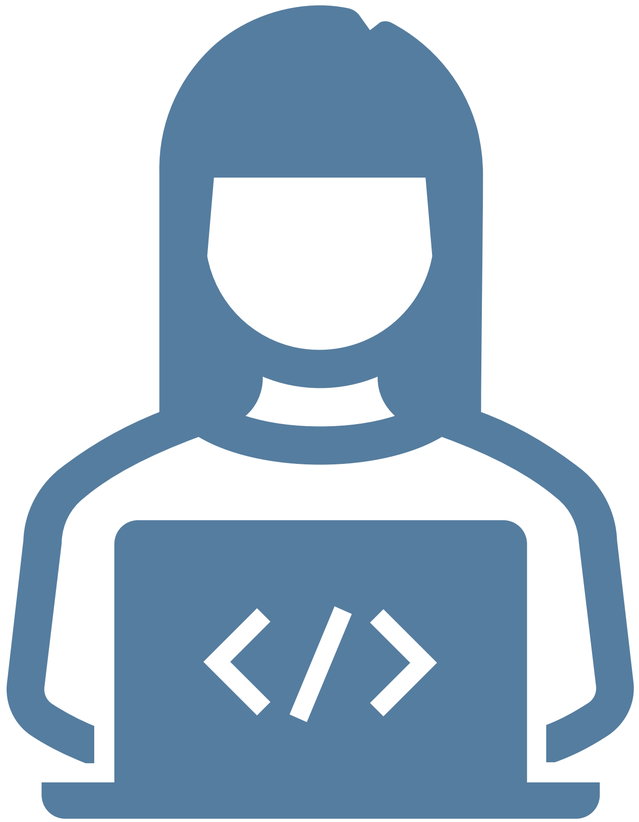
```
# no brain acrobatics

x %>% f(y) %>% g(z)
```

**pipe symbol**

FROM EXCEL TO R

# LIVE CODING 2 – TIDYVERSE



https://www.tidyverse.org/

# TIDYVERSE CHEAT SHEET

*readr, tidyr, dplyr, …*

## Read Data (*readr*)

**Reading tabular data**
There are solutions for multiple data types

```
read_excel()  # using readxl package
read_table()
read_csv()
```

**Useful arguments**
Skip lines: `read_csv(file, skip=1)`
Read subset: `read_csv(file, n_max=1)`

**Data types**
readr guesses the types of each column and tells you about it

```
("Parsed with column specifications: …")
```

## Workflow

**Tidyverse workflow**

```
df %>%
   select(col1)

df %>%
   filter(col1 < x)

df %>%
   summarize(n_1 = n(),
             avg_1 = mean(col1),
             sd_1 = sd(col1))
```

**Select column**

```
df$col1
```

## Data Manipulation (*dplyr*)

**Summary**

```
summarize()
count()
```

**Group**

```
group_by()
```

Functions will manipulate each group separately and combine results.

**Extract and sort observations (rows)**

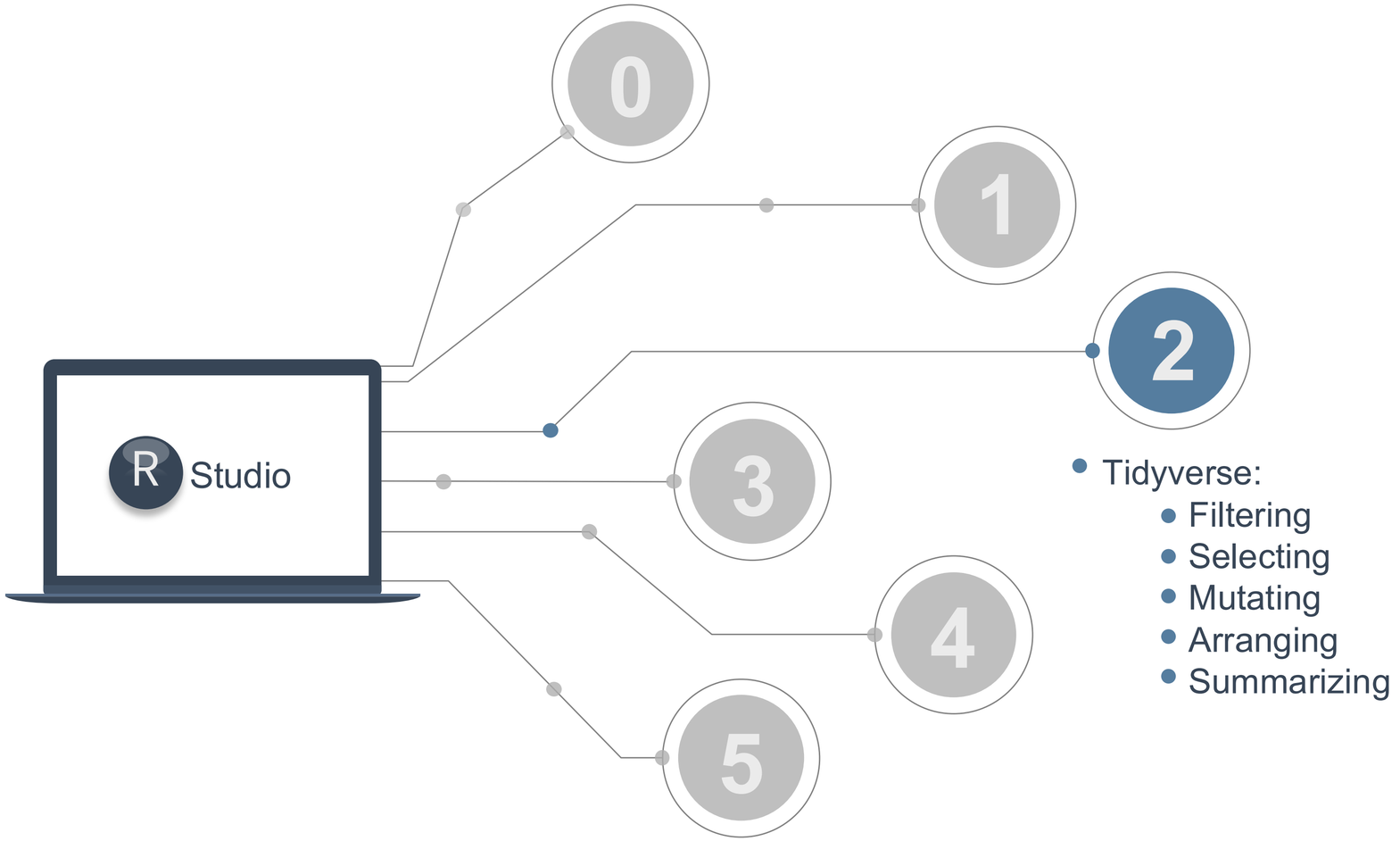
```
filter()  # subset rows by condition
distinct() # subset to unique values
top_n()  # subset by position
arrange() # sort low->high, other way with desc()
```

**Manipulate variables (columns)**

```
select()  # subset rows by condition
mutate(new_name = f(column))
mutate(new_name = ifelse(col1 < x, "Yes", "No"))
mutate(new_name = col1 + col2)
```

## HELP

R Documentation (e.g. enter **`?dplyr::filter`** and see examples)
Much more info and detailed cheat sheets:
https://brianward1428.medium.com/introduction-to-tidyverse-7b3dbf2337d5
It also helps to google "*tidyverse + whatever_you_want_to_do*"




- Tidyverse:
  - Filtering
  - Selecting
  - Mutating
  - Arranging
  - Summarizing

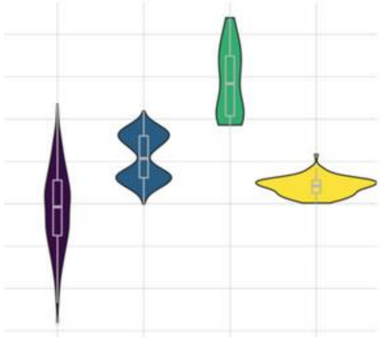
TIDYVERSE

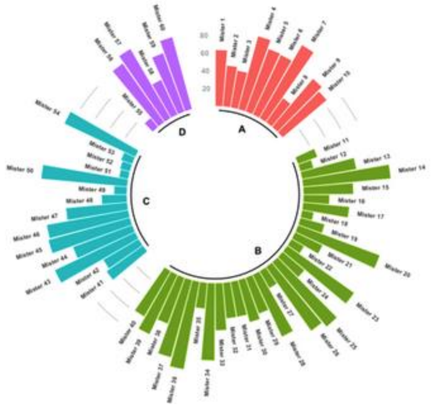
# EXERCISE **2**

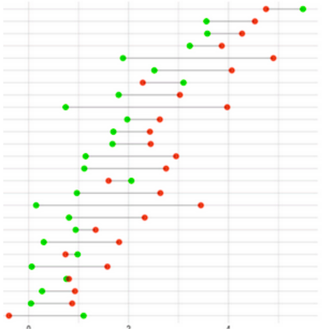
https://www.r-graph-gallery.com/

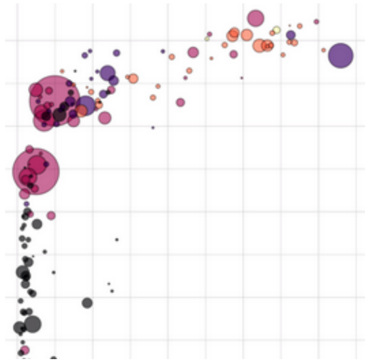
# GGPLOT2 - EASY GRAPHICS

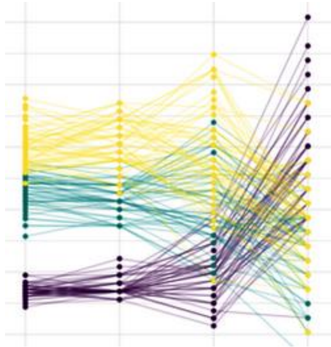
- Aesthetically pleasing graphics
- Well-defined “additive” (+) structure
- Integrates perfectly with tidy data
- Great documentation & community

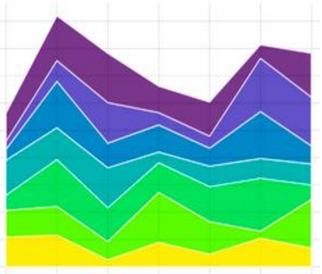












http://tutorials.iq.harvard.edu/R/Rgraphics/Rgraphics.html

# GGPLOT2
# ADDITIVE STRUCTURE




```
ggplot(data,
       aes(X, Y,
       color, fill, size, group, alpha..)) +
  geom_line()
```

**Aesthetics**

Mapping variables to groups or magnitudes - color, fill, group (all grouping variable), size, shape, etc.

**Geoms**

geom_point()
geom_boxplot()
geom_histogram()
Inherit global aesthetics, Overlay

http://tutorials.iq.harvard.edu/R/Rgraphics/Rgraphics.html

# GGPLOT2
# ADDITIVE STRUCTURE

DATASET

AESTHETICS

GEOMETRIES

FACETS

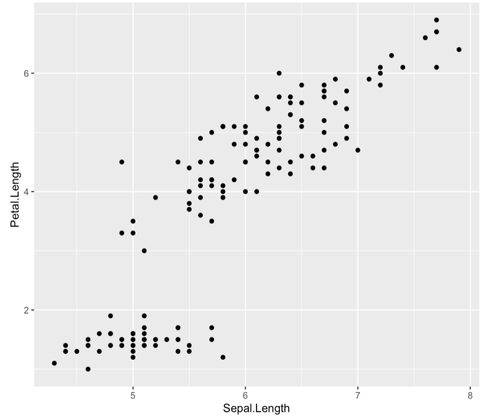
SCALES

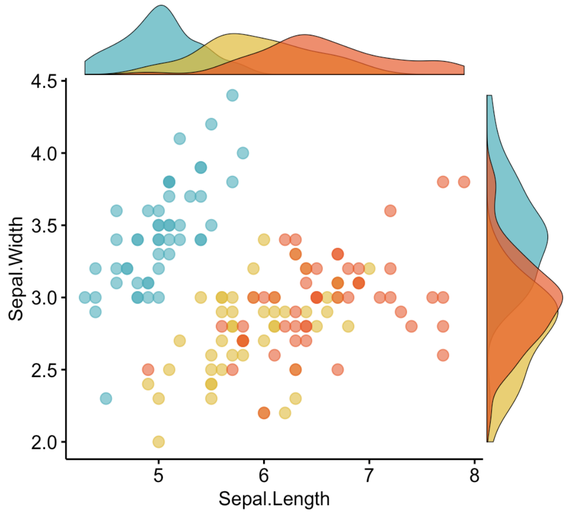
STATISTICS

THEME+LABS



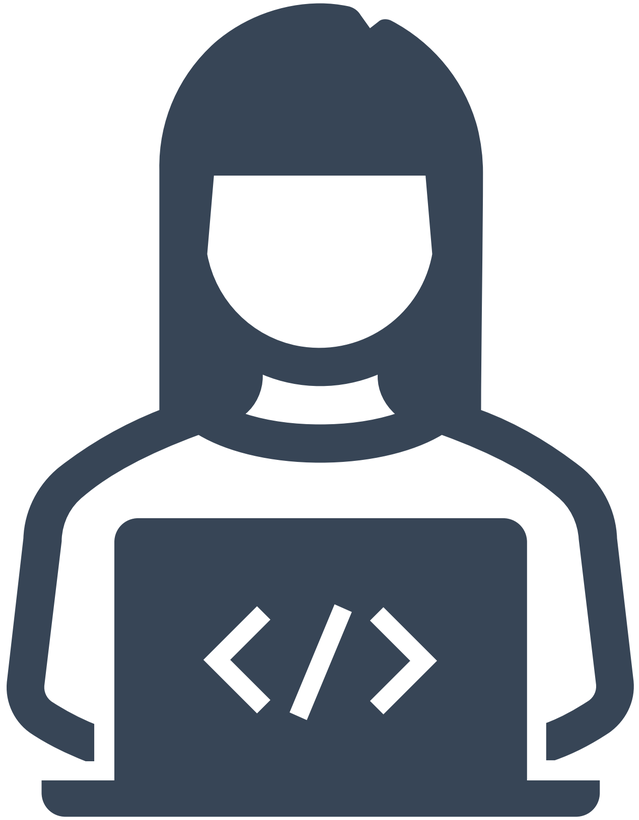
```
ggplot(data,
       aes(X, Y,
       color, fill, size, group, alpha..)) +
  geom_line()
```



FROM EXCEL TO R

# LIVE CODING 3 – GGPLOT 2



https://rstudio.com/wp-content/uploads/2016/11/ggplot2-cheatsheet-2.1.pdf

# GGPLOT CHEAT SHEET

## Define Plot:

```
ggplot(data = my.data,
       aes(x = x.var,
           y = y.var))
```

## Add Plot Type:

```
… + geom_point() # scatter plot
… + geom_line()
… + geom_boxplot()
… + geom_col()
… + geom_density()
… + geom_histogram()
```

GET STARTED

## One Color:

```
ggplot(…, aes(…),
color = "green")
```

## Color Fill by Group:

```
ggplot(…, aes(…,
fill = z.var))
```

## Custom Colors:

```
… + scale_*_manual(values = c())
ex: scale_color_manual(values = c("blue", "pink"))
```

COLORS

## More Colors:

```
… + scale_fill_grey(start = 0.2, end = 0.8)
… + scale_fill_gradient(low="white", high="red")
```

## Labels:

```
… + labs(title = ”Title”,
         x = ”X label”,
         y = ”Y label”)
```

COLOR SCALES & THEMES

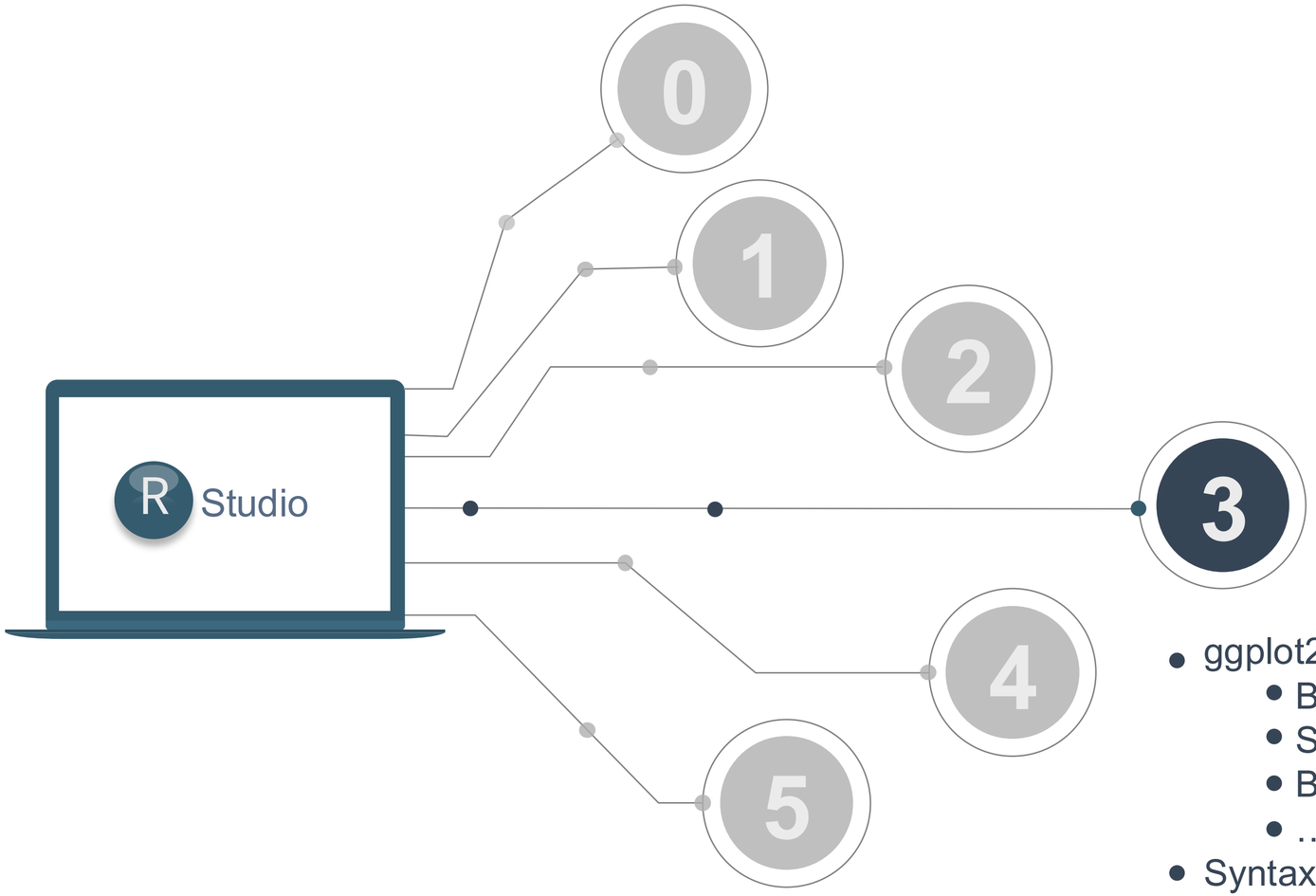
## Theme:

```
… + theme_bw()
… + theme_minimal()
… + theme_dark()
… + theme_classic()
```

## Text:

```
… + theme(legend.position = *)
* = ”none”, ”top”, ”bottom”, ”left”, ”rigth”
```

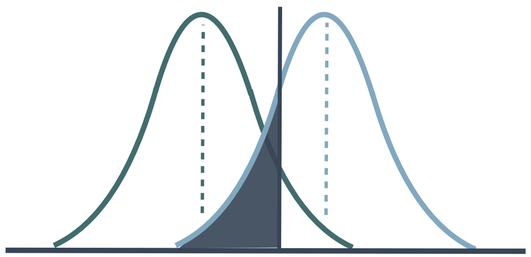
TEXT

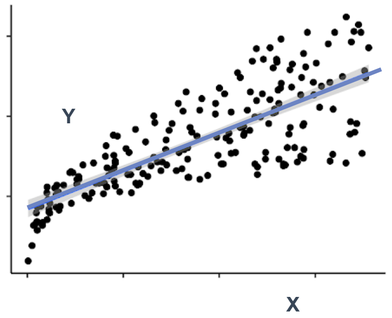


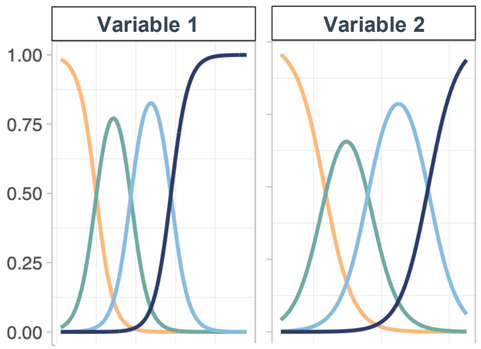

- ggplot2
  - Bar plot
  - Scatter plot
  - Boxplot
  - …
- Syntax
- Aesthetics

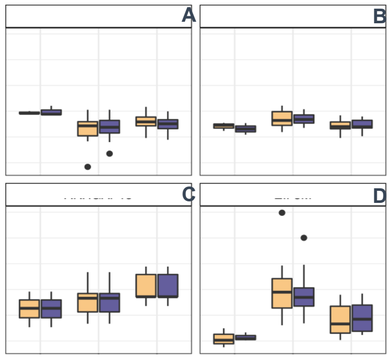
GGPLOT2

# EXERCISE 3






http://www.sthda.com/english/

# R - A STATISTICAL SCRIPTING LANGUAGE

**MODEL FUNCTIONS**

`lm()`, `glm()`
`lmer()`, `glmer()`,
`nls()`, …

**EMMEANS PACKAGE**

`emmeans()`,
`pairs()`, `cld()`

**APPLY TO MODEL**

`summary()`, `anova()`,
`confint()`, `predict()`,
`drop1()`, `update()`,
`step()`, …

**MORE FUNCTIONS**

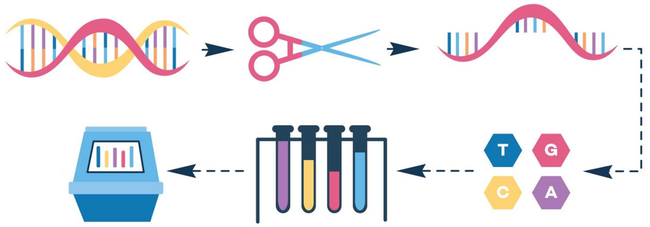
`t.test()`, `cor()`,
`cor.test()`, `aov()`,
`quantile()`,
`p.adjust()`,
`rank()`, …

# Let’s use R in a statistical analysis



**YOU**

The researcher with R skills!



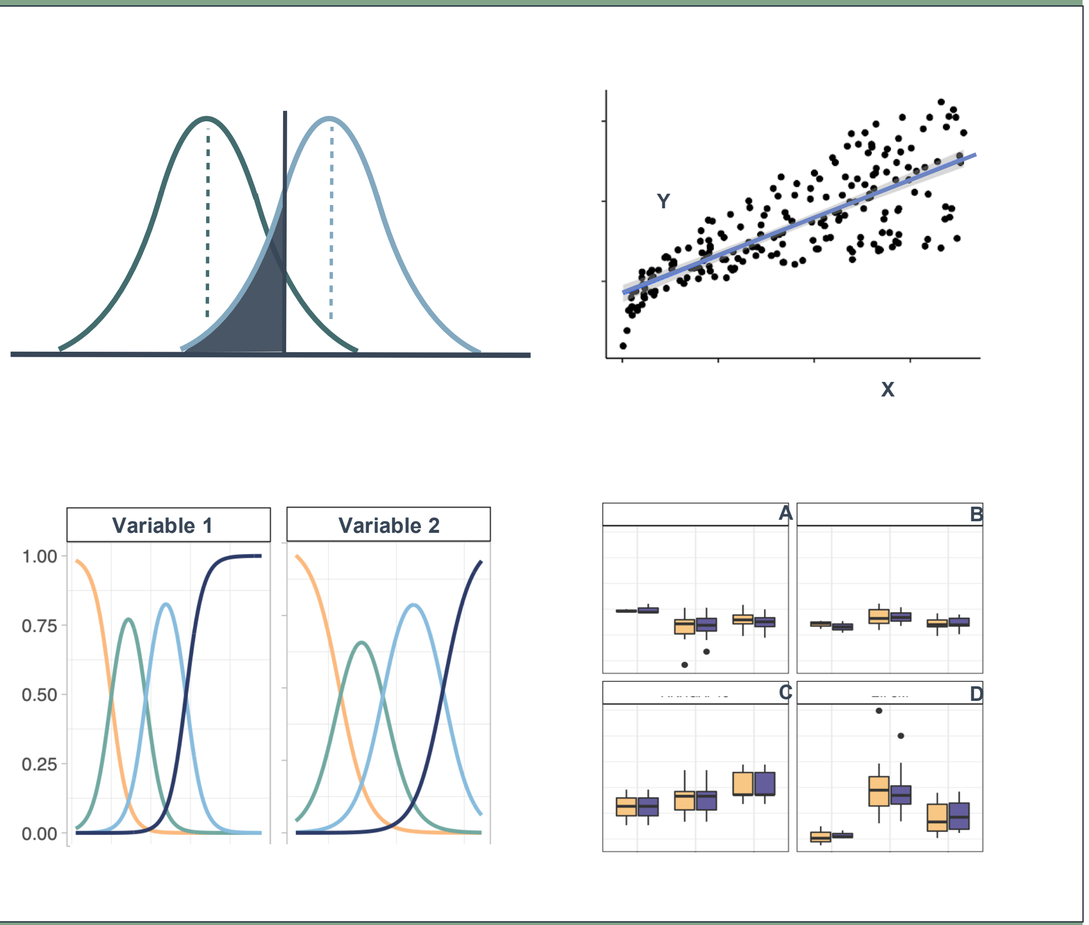

**YOUR DATA**

Gene expression data, i.e. how much a gene is used to produce a protein. Multiple genes measured in one experiment.



**YOUR DISEASE OF INTEREST**

Psoriasis, an immune-mediated skin disorder.




http://www.sthda.com/english/

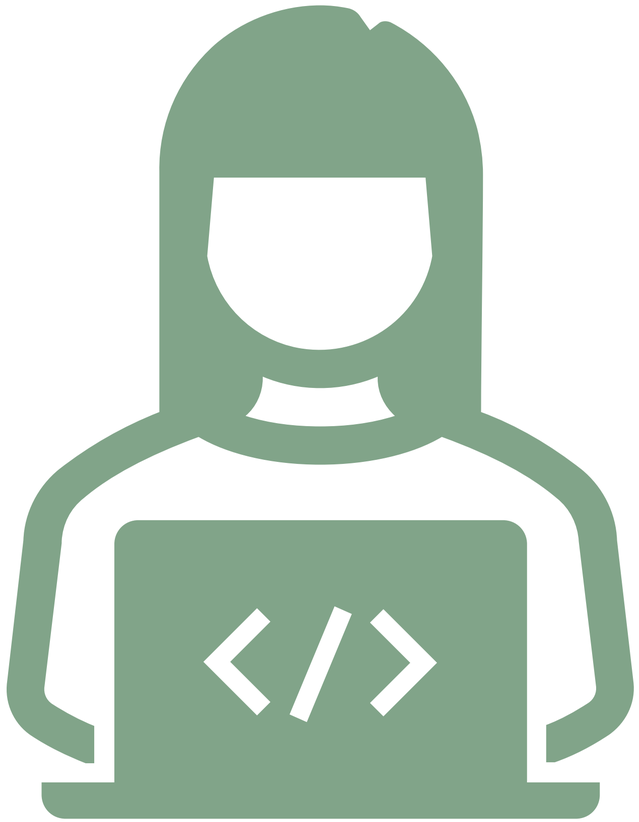
# **R** - A STATISTICAL SCRIPTING LANGUAGE

**During this session:**

- *Cooperatively* discuss and share ideas about the data
- Apply steps of basic statistical analysis for hypothesis testing consistent with the given data
- Suggest conclusions based on your analysis, regarding the association between psoriasis and gene expression levels

FROM EXCEL TO R

# LIVE CODING 4 – APPLIED STATS



https://rstudio.com/resources/cheatsheets/ for various relevant cheat sheets.
Other example: https://www.dummies.com/programming/r/statistical-analysis-with-r-for-dummies-cheat-sheet/

# STATS CHEAT SHEET

## GET STARTED

**Import Data:**

```
read_excel("my.data.xlsx")
```

**Overview of Data:**

```
summary(my.data)
nrow(my.data)
length(my.data)
names(my.data)
```

## REGRESSION MODELS

**Linear:**

```
lm(y~x, data=my.data)
confint(model)
```

**Logistic:**

```
glm(y~x, data=my.data)
```

**Linear Mixed:**

```
lmer(y~x + (1|z), data=my.data)
```

**Check Model:**

```
summary(model)
par(mfrow=c(2,2))
plot(model)
```

## TESTS/ COMPARISONS

**ANOVA:**

```
anova(model2, model1)
```

**F-Test:**

```
drop1(model, test="F")
```

**Emmeans:**

```
emmeans(model,~x)
pairs(emmeans(model,~x))
predict(model, newdata)
```

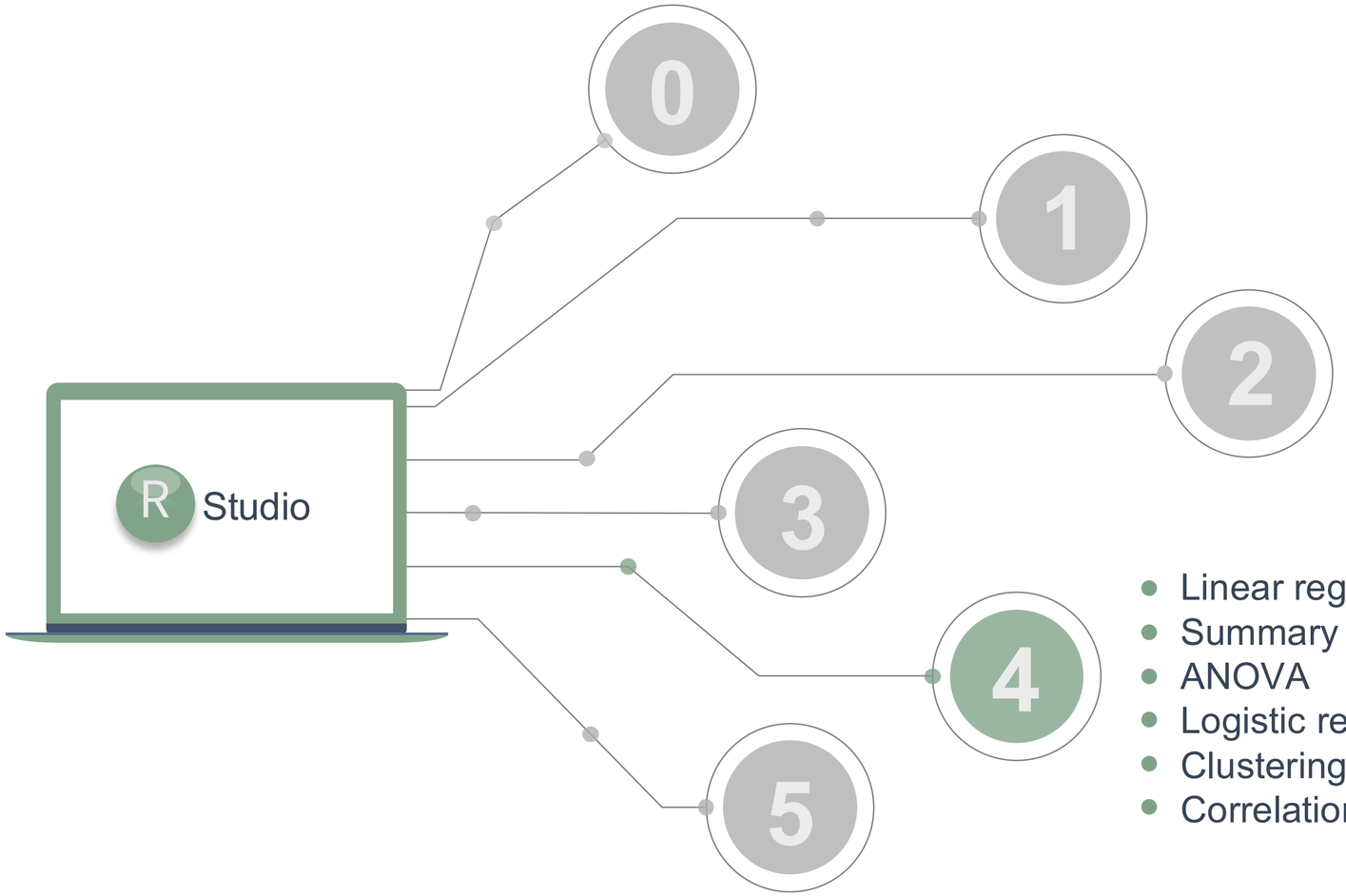
## VARIABLES

**Check Type:**

```
table(my.data$x)
is.numeric(my.data$x)
is.factor(my.data$x)
```

**Change Type:**

```
mutate(my.data, x = factor(x))
fct_collapse(x)
```




- Linear regression
- Summary Statistics
- ANOVA
- Logistic regression
- Clustering
- Correlation

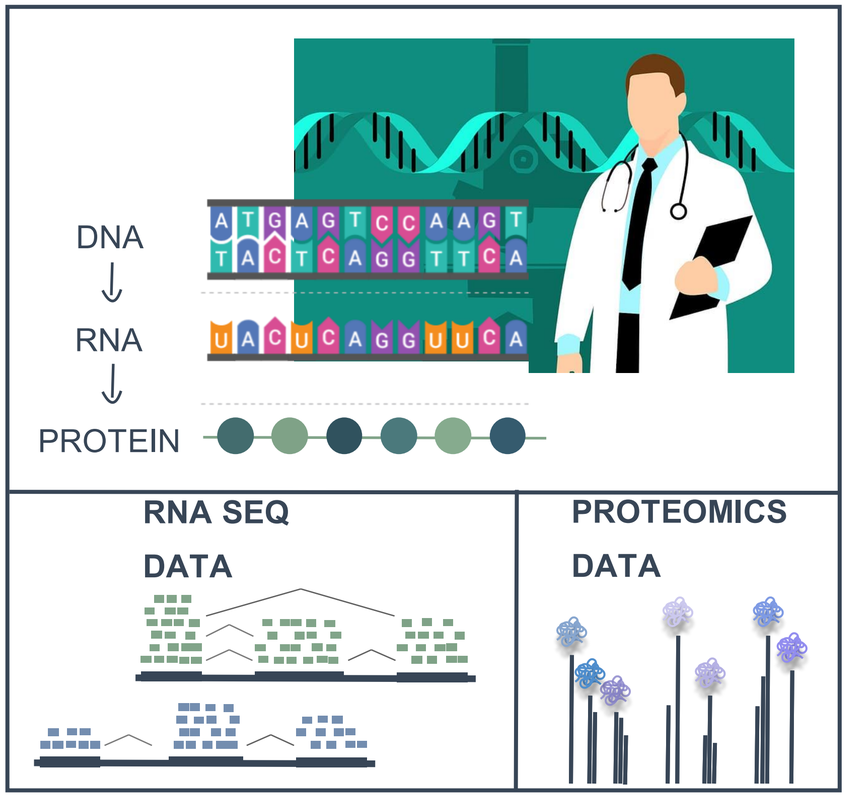
Statistics in R

# EXERCISE **4**

https://www.r-graph-gallery.com/

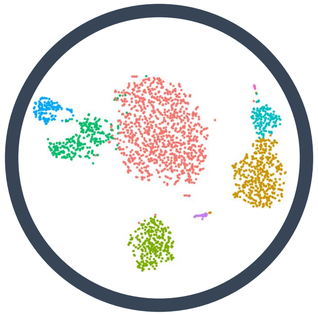
# BIOINFORMATICS IN **R**

## HIGH THROUGHPUT DATA

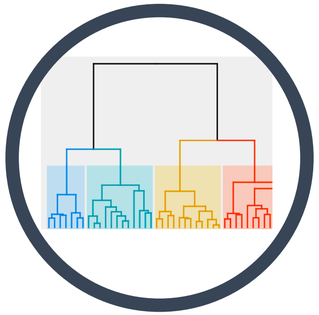



## BIOINFORMATIC ANALYSIS

### DIMENSIONALITY REDUCTION



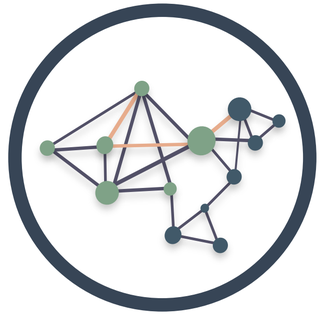
### CLUSTERING

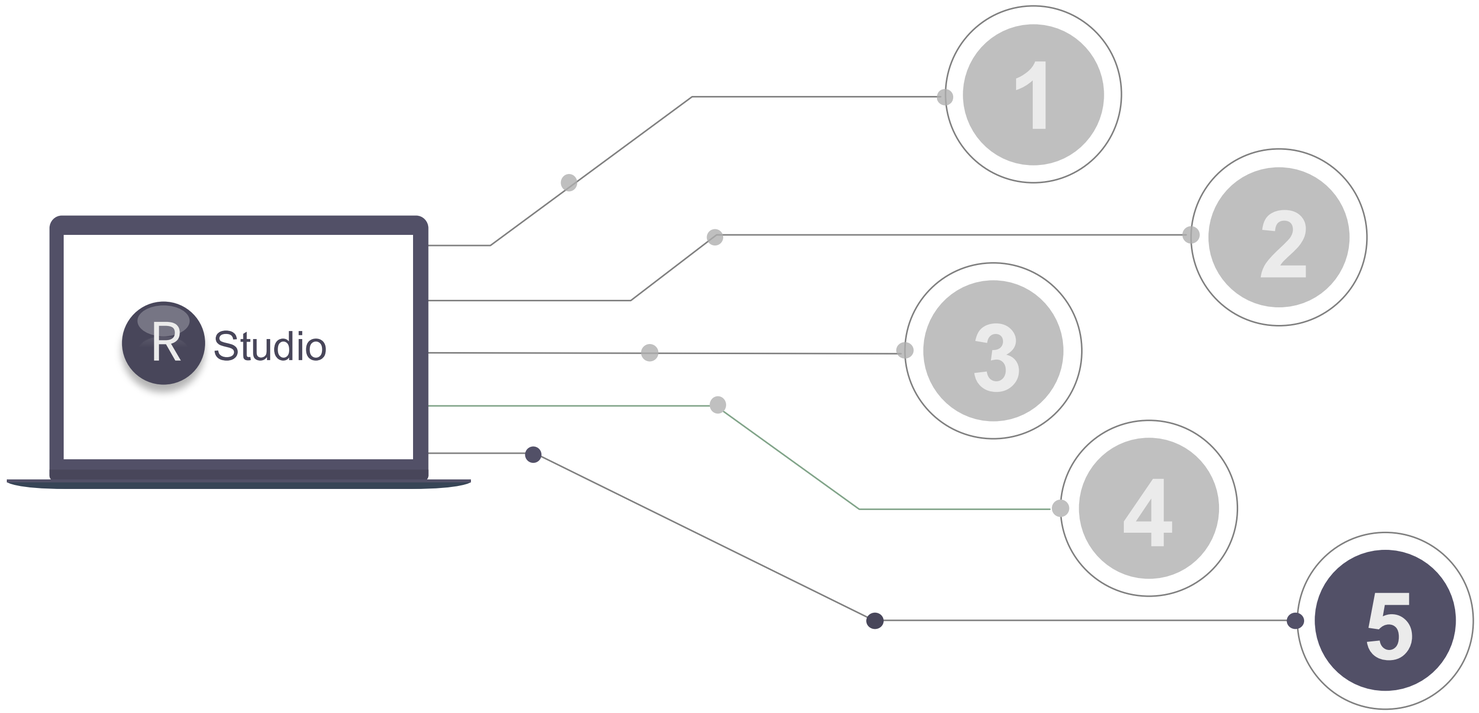


### MACHINE LEARNING



### NETWORK ANALYSIS






- Tidyverse
- ggplot
- Stats
- Quarto

Statistics in R

# EXERCISE **5**

# THE TOP OF THE R ICEBERG

## STATISTICAL ANALYSIS

Statistical models (linear, generalized, mixed, …)

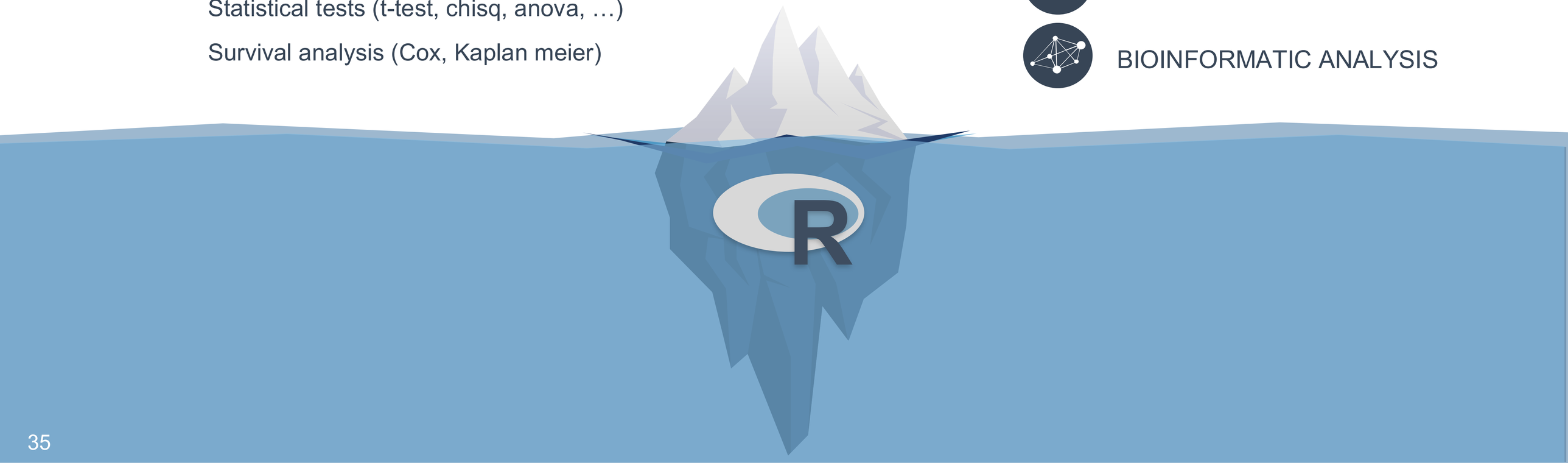
Statistical tests (t-test, chisq, anova, …)

Survival analysis (Cox, Kaplan meier)

## DATA MANGEMENT

## EASY PLOTTING

## BIOINFORMATIC ANALYSIS



# WANT MO-R-E?

The Section for Biostatistics offers a number statistics-oriented R courses:

https://publichealth.ku.dk/about-the-department/biostat/

**Spring**

**Basic Statistics for Health Researchers (Danish course)**
ECTS: 9,0

**Epidemiological methods in medical research**
ECTS: 7

**Advanced topics in health research B**
ECTS: 2,8

**Statistical methods in bioinformatics**
ECTS: 3,5

**Statistical analysis of survival data**
ECTS: 4,9

**Programming and statistical modelling in R**
ECTS: 1,6

**Bayesian methods in biomedical research**
ECTS: 2,4

**Psychometric validation of patient reported outcome measures**
ECTS: 2,5

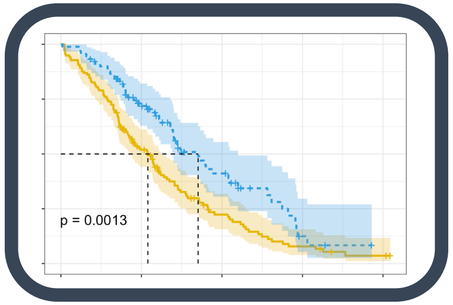
**Introduction to validation of patient reported outcome measures**
**Causal inference I**
ECTS: 2,5

**Use of the statistical software R**
ECTS: 2,1

* These are screenshots. Go to the website and scroll down to ‘Teaching’

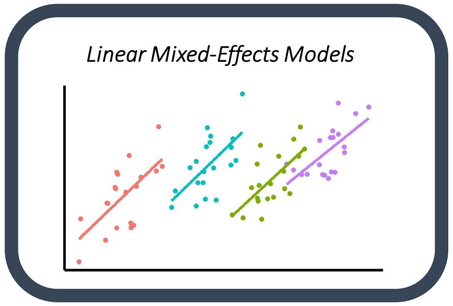
TEASER

# STATISTICS in **R**




## Survival Analysis

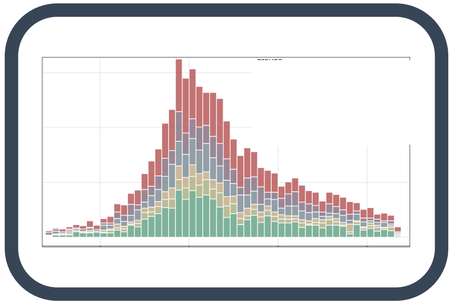
**survival**: https://rviews.rstudio.com/2017/09/25/survival-analysis-with-r/
**survminer:** https://cran.r-project.org/web/packages/survminer/survminer.pdf (https://rpkgs.datanovia.com/survminer/ )




## Mixed-Effects Models

**lme4:** https://cran.r-project.org/web/packages/lme4/vignettes/lmer.pdf

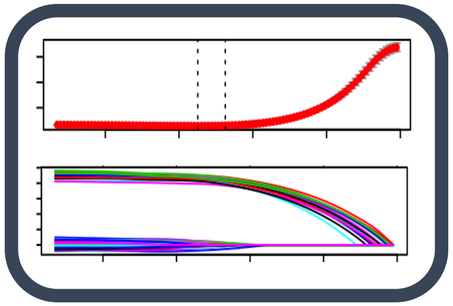
https://cran.microsoft.com/snapshot/2017-08-01/web/packages/sjPlot/vignettes/sjplmer.html
**glmmTMB**: https://cran.r-project.org/web/packages/glmmTMB/index.html



## Epidemiological Analysis

**Epi:** https://cran.r-project.org/web/packages/Epi/index.html

**pubh:** https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/
https://cran.r-project.org/web/packages/incidence/vignettes/customize_plot.html
https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/



## Elastic-Net Regression

## (R

**glmnet:** https://cran.r-project.org/web/packages/glmnet/glmnet.pdf

**elasticnet:** https://cran.r-project.org/web/packages/elasticnet/elasticnet.pdf

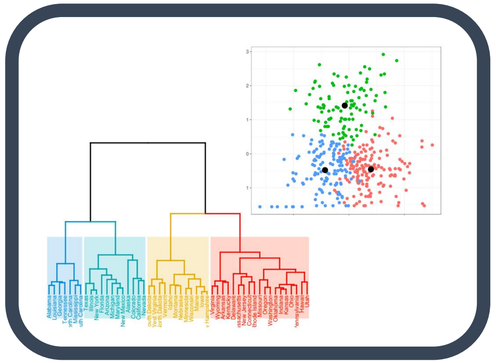
https://www.datacamp.com/community/tutorials/tutorial-ridge-lasso-elastic-net

TEASER

# Machine Learning

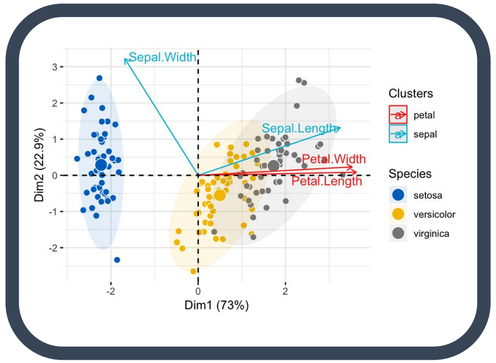
https://lgatto.github.io/IntroMachineLearningWithR/an-introduction-to-machine-learning-with-r.html

## Clustering



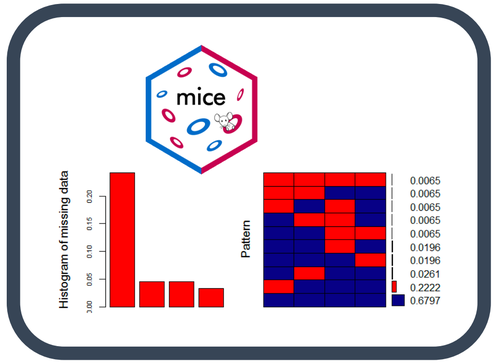
https://statsandr.com/blog/clustering-analysis-k-means-and-hierarchical-clustering-by-hand-and-in-r/

## Feature Selection: PCA



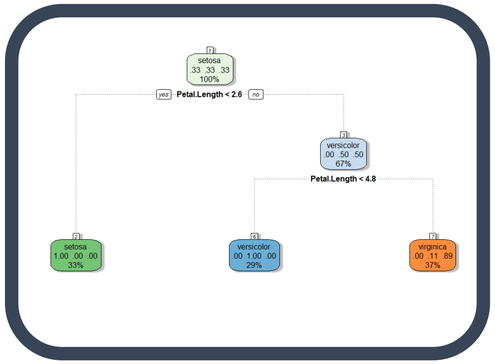

https://bioconductor.org/packages/release/bioc/vignettes/PCAtools/inst/doc/PCAtools.html

## Missing Data



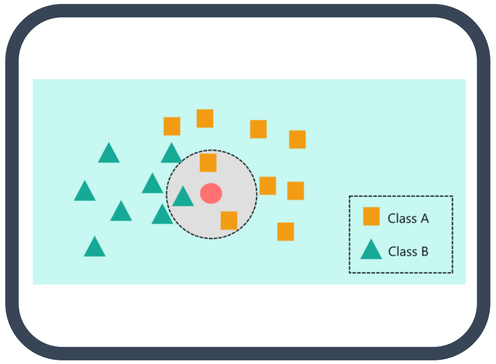

https://amices.org/mice/
https://datascienceplus.com/imputing-missing-data-with-r-mice-package/

## Random Forest



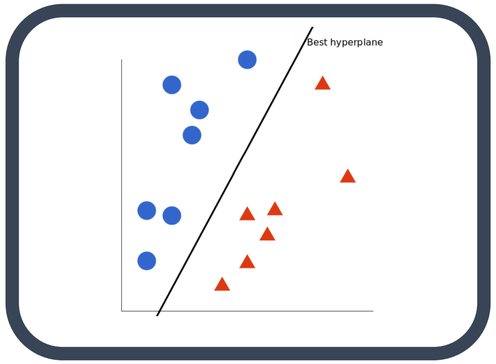

https://www.blopig.com/blog/2017/04/a-very-basic-introduction-to-random-forests-using-r/

## kNN



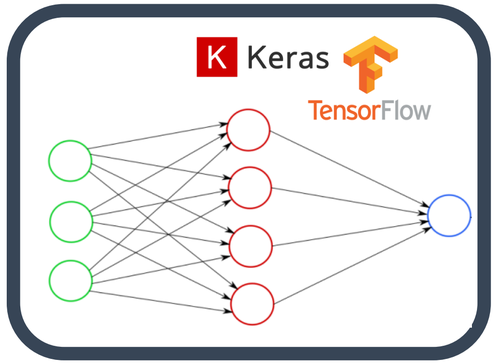

https://www.edureka.co/blog/knn-algorithm-in-r/

## SVM



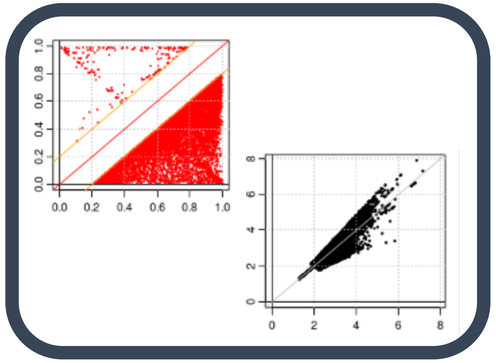

https://cran.r-project.org/web/packages/e1071/vignettes/svmdoc.pdf

## Neural Networks



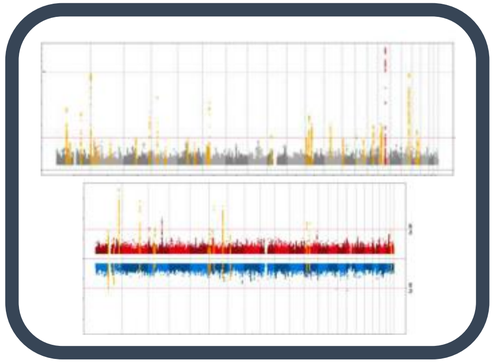

https://keras.rstudio.com/
https://tensorflow.rstudio.com/

## GWAS - QC & Data Harmonization



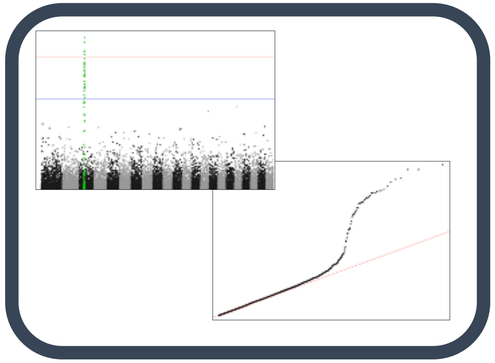
**EasyQC**: https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/

## GWAS Data Management & Plots



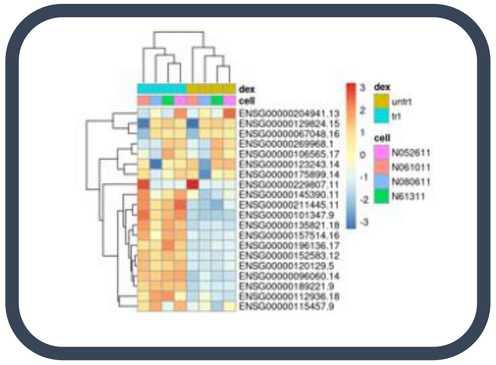
**EasyStrata**: https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/

## More Plotting…



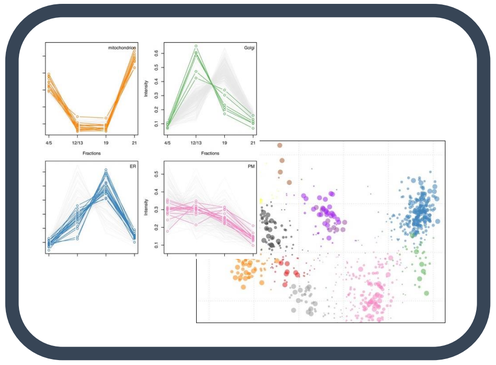
**Manhattan and QQ plots**: https://cran.r-project.org/web/packages/qqman/vignettes/qqman.htm l

## Gene Expression Analysis



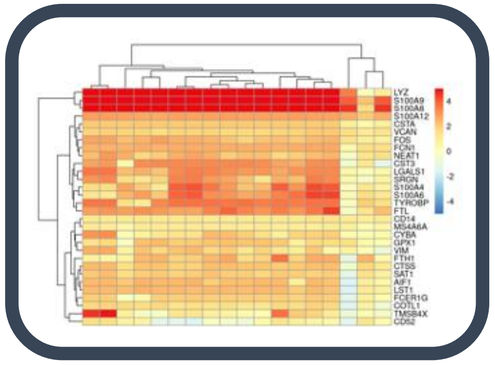

**DESeq2, limma, EdgeR, etc.:** http://www.bioconductor.org/packages/release/BiocViews.html#___RNASeq

## Proteomics Analysis



**RforProteomics:** https://www.bioconductor.org/packages/release/BiocViews.html#___Proteomics

## Single-Cell RNASeq



https://cran.r-project.org/web/packages/e1071/vignettes/svmdoc.pdf

TEASER

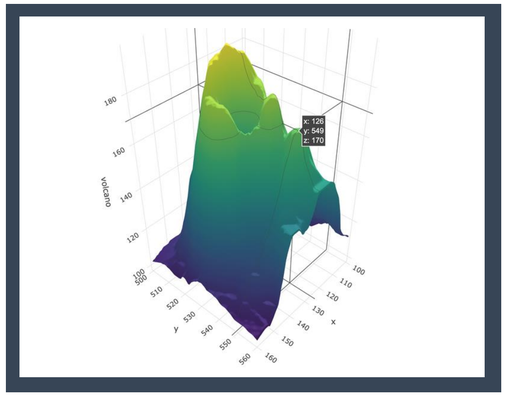
# Omics Data

http://www.bioconductor.org/packages/release/BiocViews.html

FROM EXCEL TO R

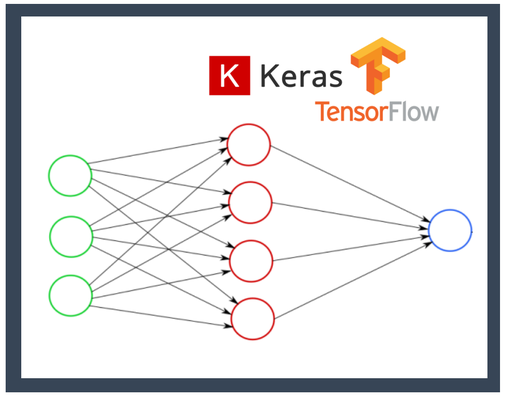
# COOL STUFF IN R

## PLOTTING IN 3D




https://plotly-r.com/d-charts.html

## DEEP LEARNING




https://keras.rstudio.com/
https://tensorflow.rstudio.com/

## BAYESIAN STATISTICS



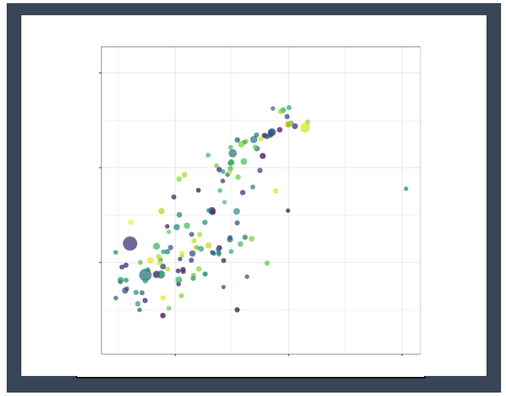

https://mc-stan.org/users/interfaces/rstan

## WEBPAGE WITH R SHINY




https://shiny.rstudio.com/

## INTERACTIVE PLOTS



https://gganimate.com/articles/gganimate.html

## MAIL AND MESSAGES




https://github.com/briandconnelly/pushoverr

FROM EXCEL TO R

# FREE CONSULTATIONS - Drop by!!!

## Experimental Design

Randomization, confounders, power estimates

## Data, workflows, and infrastructure

Compliance (GDPR, *Projects Like Mine*);
HPC & Storage,
Reproducibility (Git, Dockers, FAIR, Pipelines)

## Data analysis & bioinformatics

R analysis, omics, single-cell, clinical & registry data
Batch effects, normalization, missing data

## Research outputs

Visualizations, Reproducible reports, dashboards and apps, Data sharing /Hosting



# EVENTS and CONFERENCES




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https://www.linkedin.com/company/ucph-heads/

**Read more on HeaDS Website!**

https://heads.ku.dk

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